

to deploy applications written in a wide range of languages and Web frameworks, using a variety of data services.

Helion Stackato helps in the automatic configuration of language runtime, Web servers, application dependencies, databases and other services. Stackato can be run in any data centre using the hypervisor of choice or by using a supported cloud hosting provider. It provides IT operators with simplified deployment and cloud native services. It is infrastructure agnostic, guaranteeing compatibility across cloud infrastructures. With the Helion Service Manager, operators can easily manage application services while leveraging the Helion Control Plane to ensure the entire application life cycle.

HPE Helion Stackato provides multi-cloud integration support for enterprise level developers by using the Web console. Using this, developers can create applications from any code repos (e.g., GitHub) or any of their enterprise grade repos, enable the full build, and then test and deploy the complete pipeline. The management of the application's life cycle has thus been made very easy. As mentioned earlier, any of the popular development languages like Java, .NET, Python, Go, Ruby and Node.js can be chosen. Also, developers can build and use plugins for the existing popular IDEs like that of MS Visual Studio, Eclipse, etc.

HPE Helion Stackato platform services: These are listed below.

Helion Control Plane (HCP): This is a core platform service that HPE Helion Stackato uses to manage service life cycles and communicate with the underlying cloud provider (IaaS) layer.

Helion Service Manager (HSM): This provides a repository of services that can be used by applications.

Helion Code Foundry (HCF): This is a Cloud Foundry certified elastic runtime that simplifies cloud native application development and hosting.

Helion Code Engine (HCE): This is a continuous delivery service that integrates with public or private Git repositories. HCE is a flexible and extensible continuous integration/continuous development (CI/CD) pipeline.

HPE Helion Stackato Console (HSC): This is a Web interface that's used to manage HCF and HCE features.

The advantages of using Stackato are:

- Stackato provides an agile, robust cloud application platform
- It automates the auto scaling of the Stackato cluster using the supported cloud dashboard
- Working seamlessly together, Stackato and the supported cloud enable you to expand and reduce cloud resources based on user demand, thus reducing the setup time
- Stackato helps in improving the productivity of cloud administrators and developers

Stackato is well suited for enterprise cloud application deployments, and also helps in the application model when compared to IaaS or cloud orchestration software. It combines the flexibility offered by direct VM access on IaaS with the highly automated configuration provided by PaaS. Computing resources are shared efficiently and securely by giving each application its own Linux container (using LXC), which can be extensively customised to suit the application it is hosting.

Stackato supports the following:

- Citrix XenServer
- OpenStack
- KVM
- VMware
- AWS
- HP Cloud services
- Dell

Cloudify

Cloudify was developed and designed on the principles of openness to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services. The latest version of Cloudify is 4.2 and it incorporates enhanced features like advanced security, control and true self-service.

Cloudify has very good orchestration support for NFV (network function virtualisation). It's TOSCA (Topology and Orchestration Specification for Cloud Applications) based, open and pluggable architecture provides end-to-end management and orchestration (MANO) of the NFV lifecycle.

This enables telecom based organisations and operators to build best-of-breed NFV stacks and reduce the overhead costs considerably.

Cloudify 4.2 introduced a totally new concept for container orchestration with Kubernetes (Kubernetes Cloud Native Orchestration). Now onwards, Cloudify will offer Kubernetes Provider. This will allow you to utilise Cloudify as a cloud provider; so you can easily deploy a cluster as well as scale and auto scale the number of nodes natively, configure networking and load balancing, and have storage and compute customisation as well as native multi-cloud support, as and when required.

Figure 3 shows the detailed architecture of Cloudify.

Cloudify also enables ARIA TOSCA integration. In order to meet the complete standards requirements, TOSCA Simple Profile 1.0 is now supported using the ARIA plugin. The plugin allows orchestrating TOSCA CSAR packages

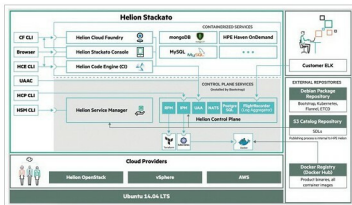


Figure 2: HPE Helion Stackato architecture (Source: <https://docs.hpcloud.com/stackato/stackato/planning/architecture.html>)